

# Paper C-03: Trade Networks: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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August 2026

*Shared Part IV notation and evidence boundaries are defined in `PART_IV_METHODS_AND_CLAIM_BOUNDARY.md`. This paper states only its local observables, comparison, and failure conditions.*

## Abstract

This paper develops a falsifiable CDFD/AFL model of Trade Networks. It maps goods, services, capital, information, and contractual demand to driving flux  $\Phi$ , ports, tariffs, logistics, finance, and network dependencies to active constraint  $C$ , substitution, inventory, rerouting, diplomacy, and production change to responsiveness  $S$ , and contracts, specialization, infrastructure, and disruption history to structural memory  $M_s$ . The target outcome is trade volume, shortages, prices, and recovery, tested against customs data, input-output tables, shipping traces, prices, and disruption records. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

## 1 Introduction

The global economy is a singular, highly integrated fluid network. Capital, technology, and raw materials naturally seek to flow from regions of surplus to regions of deficit, driven by the thermodynamic gradient of profit and necessity.

However, nations frequently intervene in this natural flow for geopolitical reasons. Tariffs, embargoes, and sanctions act as artificial dams placed across the river of global commerce. In CDFD, a sovereign state implementing a tariff is actively manipulating the topological constraints of the global active surface.

## 2 The Geoeconomic Adaptive Surface

We govern bilateral and global trade using the Adaptive Operating Ratio:

$$\Psi_s = \left( \frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

For a specific trade corridor between two nations:

- **Flux** ( $\Phi$ ): The monetary value and physical volume of exports and imports crossing the border.
- **Constraint** ( $C$ ): Tariffs, quotas, regulatory misalignment, and outright economic sanctions.
- **Responsiveness** ( $S$ ): The elasticity of the domestic economy to substitute imports, find alternative trading partners, or pivot domestic production.
- **Memory** ( $M_s$ ): Decades-long trade agreements, entrenched supply chains, and specialized infrastructure (e.g., specific pipelines or localized shipping hubs). High memory locks the economy into dependency on a specific partner.

### 2.1 Trade Wars and Topological Severing

In a free-trade era, tariffs are low, and  $\Phi$  is extremely high. The global system is highly integrated and operates in a sustained growth regime ( $\Psi_s > 1$ ).

If a trade war begins, nation A applies a massive tariff against nation B. The constraint  $C$  spikes artificially. If nation B's economy is highly specialized and dependent on this specific corridor (high memory  $M_s$ ), it cannot easily pivot. The adaptive ratio for that specific trade sector crashes to  $\Psi_s \ll 1$ .

The flow  $\Phi$  of critical components ceases. The sanctioned or tariffed nation enters an ischemic economic state. To survive, the nation must drastically increase its internal responsiveness  $S$ —forcing expensive domestic production or forging entirely new, inefficient trade corridors with secondary nations. The artificial constraint forces a massive, thermodynamically inefficient topological rewiring of the global surface.

## 3 Measurement Layer

## 4 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

### 4.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Trade Networks, the proposed drive is goods, services, capital, information, and contractual demand. The active limitation is ports, tariffs, logistics, finance, and network dependencies. Responsiveness is carried by substitution, inventory, rerouting, diplomacy, and production change, while retained state is represented by contracts, specialization, infrastructure, and disruption history. The outcome to explain is trade volume, shortages, prices, and recovery.

Table 1: Operational CDFD/AFL map for Paper C-03.

| Term   | Paper-specific observable meaning                                    | Measurement question                                                      |
|--------|----------------------------------------------------------------------|---------------------------------------------------------------------------|
| $\Phi$ | goods, services, capital, information, and contractual demand        | What rate, load, gradient, or demand enters the focal system?             |
| $C$    | ports, tariffs, logistics, finance, and network dependencies         | Which measured bottleneck limits transfer, function, or recovery?         |
| $S$    | substitution, inventory, rerouting, diplomacy, and production change | Which process changes effective capacity or reroutes the load?            |
| $M_s$  | contracts, specialization, infrastructure, and disruption history    | Which retained state makes equal present loads produce different futures? |
| $Y$    | trade volume, shortages, prices, and recovery                        | Which outcome is predicted out of sample, with uncertainty?               |

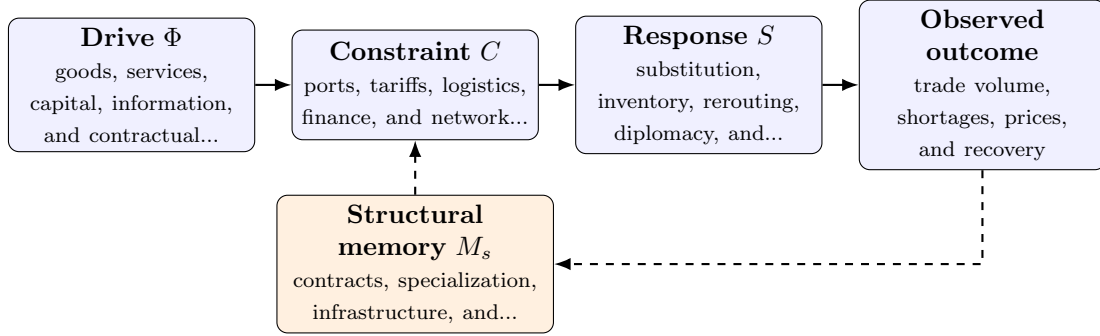


Figure 1: Paper C-03 observable CDFD/AFL architecture for Trade Networks. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

## 4.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in goods, services, capital, information, and contractual demand arrives faster than substitution, inventory, rerouting, diplomacy, and production change can compensate. This raises or redistributes ports, tariffs, logistics, finance, and network dependencies. If the load persists, the retained state encoded by contracts, specialization, infrastructure, and disruption history changes the next response, so a later event of equal magnitude can produce a different trajectory in trade volume, shortages, prices, and recovery. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

## 4.3 Cross-Scale Translation Without Mechanistic Erasure

For the socioeconomic systems family, the cross-domain translation compares human and institutional throughput, unequal access to capacity, adaptive coordination, and historical path dependence. The variables are not morally neutral: aggregation can hide who receives flow, who bears constraint, and whose prior losses become persistent memory. [1, 4, 2, 3] The required scale declaration for this paper is days to decades; firms to the world economy. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

## 4.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper C-03. Each rung can fail independently.

| Test                | Design                                                                                                                                                                    | Discriminating result                                                                                                |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Proxy audit         | Measure customs data, input-output tables, shipping traces, prices, and disruption records and preregister how each record maps to $\Phi$ , $C$ , $S$ , $M_s$ , and $Y$ . | The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.                         |
| Load ramp           | Apply or observe a corridor closure or supplier loss during high demand while estimating the dominant constraint and response time.                                       | A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint. |
| Recovery and memory | Match present load across units with different histories, then compare recovery paths.                                                                                    | The memory claim requires history-dependent divergence after current conditions are controlled.                      |
| Model comparison    | Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.                                                                      | The paper’s stated falsifier is: network memory does not improve shortage or trade-recovery prediction.              |

## 4.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from customs data, input-output tables, shipping traces, prices, and disruption records and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of trade volume, shortages, prices, and recovery. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

## 5 Conclusion

Trade wars and sanctions are not merely political tools; they are the intentional weaponization of thermodynamic constraints. By framing international trade within CDFD, we see that global commerce behaves as an adaptive biological membrane. Artificial geopolitical constraints force traumatic structural adaptations, supporting the testable hypothesis that economic isolation can drive either violent internal restructuring or systemic decay.

## Conflict of Interest

The author declares no conflict of interest.

## Data Availability

Part-specific runtime diagnostics are under each Part’s `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: Part\_E\_Synthesis/outputs/.

Figures: Part\_E\_Synthesis/figures/.

## 6 Reference Layer

### References

- [1] Albert-Laszlo Barabasi and Reka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999. DOI: <https://doi.org/10.1126/science.286.5439.509>.
- [2] C. S. Holling. Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4:1–23, 1973. DOI: <https://doi.org/10.1146/annurev.es.04.110173.000245>.
- [3] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. DOI: <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [4] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of small-world networks. *Nature*, 393:440–442, 1998. DOI: <https://doi.org/10.1038/30918>.